

RAJAN CHAVADA

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EDUCATION

University of Western Ontario

B.Sc. Computer Science, GPA: 3.7/4.0

Sept 2022 – Expected May 2027

London ON

PROFESSIONAL EXPERIENCE

ML/AI & Infrastructure Scientist

Borealis AI | Royal Bank of Canada

Dec 2025 – Present

Toronto, ON

- 🔧 **Led migration of 3 mission-critical credit capacity ML models** to a new NVIDIA DGX A100/H100 OCP cluster by reconfiguring Triton inference endpoints, K8 deployments, and CI/CD pipelines, **resulting in zero-downtime cutover and 99.95% production uptime during cluster deprecation.**
- 🔧 **Reduced P95 inference latency by 25% and increased throughput 2×** by optimizing GPU-based model serving (dynamic batching, quantization, container tuning) and conducting structured load/performance testing prior to release.
- 🔧 **Improved deployment velocity by 60%** by automating model build, validation, and OpenShift rollout workflows using GitHub Actions, YAML configurations, and containerized PyTorch/TensorFlow services, **cutting release cycles from days to hours.**
- 🔧 **Enhanced production reliability and security** by implementing mTLS sidecar configurations and real-time monitoring for SLA enforcement (99.9%+), **reducing incident detection time by 40% and ensuring encrypted service-to-service communication.**

Cloud Engineer

Intact Insurance

Sept 2025 – Dec 2025

Toronto, ON

- 🔧 Reduced manual infrastructure provisioning time by **65%** by designing Terraform modules for **190+** AWS accounts, enabling self-service infrastructure-as-code and accelerating ML workload deployment.
- 🔧 Built ETL pipelines that ingested infrastructure telemetry and system logs into **Amazon Bedrock** (generative AI service), enabling autonomous insights and recommendations for cloud optimization and compliance checks.
- 🔧 Deployed vulnerability management automation across **50+** Kubernetes/OpenShift clusters via DaemonSet, achieving **95%** compliance coverage and cutting security audit time by **75%** using SageMaker-based anomaly detection and automated remediation workflows.

Quantitative Software/AI Engineer

Royal Bank of Canada | Amplify

May 2025 – Aug 2025

Toronto, ON

- 🔧 Designed and shipped a multi-agent research assistant over Snowflake using Python (FastAPI) microservices and well-defined APIs, improving research efficiency by **60%** for **10,000+** daily internal queries and integrating multiple LLM tools.
- 🔧 Built Kafka-based data pipelines processing **50k+** daily articles with automated performance monitoring, maintaining sub-second latency for **18,000+** users in a large distributed environment, and feeding downstream ML ranking models.
- 🔧 Reduced CI/CD deployment time by **50%** by containerizing services with multi-stage Docker builds and Tekton pipelines across 8 Kubernetes/OpenShift clusters, accelerating safe rollout of new ML and backend components.
- 🔧 Developed and optimized forecasting services (Prophet/XGBoost) to achieve **3x** higher throughput while preserving accuracy, balancing performance, cost, and maintainability for batch inference workloads.

Full-Stack Software Engineer Intern (Client Services)

Royal Bank of Canada | Capital Markets

May 2024 – Aug 2024

Toronto, ON

- 🔧 Built a client-facing **AI chatbot** leveraging RBC's internal LLM gateway and a **vector search database** (semantic embeddings) to provide RAG-powered responses to investor service inquiries, reducing support tickets by **22%**.
- 🔧 Implemented semantic search and **vector-optimized queries** over investor portal documentation, enabling the chatbot to retrieve contextually relevant answers with high precision for common client questions about trading platforms and account management.
- 🔧 Built tooling and performance monitoring for the React-based trading platform serving **4,500** DAU, using Lighthouse CI to profile **12+** configs and catch performance regressions before release.

Full-Stack Software Engineer Intern (Global Equities)

Royal Bank of Canada | Capital Markets

May 2023 – Aug 2023

Toronto, ON

- 🔧 Engineered **internal CVE triage platform** with **React.js**, **Flask REST API**, and weekly cron scheduling parsing **P1 CVE** data from Microsoft SSMS across **6+ sources**, cutting production errors by 30% and developer triage time by **4 hours ~40%**.
- 🔧 Designed priority based caching routing layer to stream **P1 CVE** to on-call engineers, reducing MTTA by **30%** and achieving **99.8% SLA** compliance with global equities trading schedule across **3** regional trading desks

PROJECTS

NVIDIA Alert Triage & Incident Response Agent | (LangChain, LangGraph, LangFuse NVIDIA-NIM, Kafka, Postgres) [Live App](#)

Present

- **Built a multi-agent LLM triage system (LangGraph + RAG + NVIDIA NIM) for Selene/SaturnV clusters**, integrating Kafka-based real-time ingestion to automate alert classification and root-cause analysis, **targeting 70–80% MTTR reduction**

PhysioPoint AR Rehab Application | (Swift UI + ARKit) [Repo](#)

Present

- **Developed an AR-based rehabilitation assistant using SwiftUI + ARKit (ARBodyTracking)** to compute real-time joint angles and validate exercise form, enabling accessible at-home therapy for elderly users and recovering athletes.

SKILLS & OTHER

Languages & Technical Skills: Python, C++, C#, Java, Go, TypeScript, SQL, Bash, FastAPI, Flask, Node.js, API, Swift Git, Rust, SQL, noSQL, SDLC

Systems & Infra: Linux/Unix, Kubernetes, OpenShift, Docker, Terraform, CI/CD (Jenkins, Tekton, GHA), Kafka, Snowflake, SQL, real-time Streaming, API

ML/Cloud: Machine Learning, data-driven decision-making, forecasting models (Prophet, XGBoost), AWS, GCP, OpenShift, Azure, CI/CD, PyTorch, Ansible